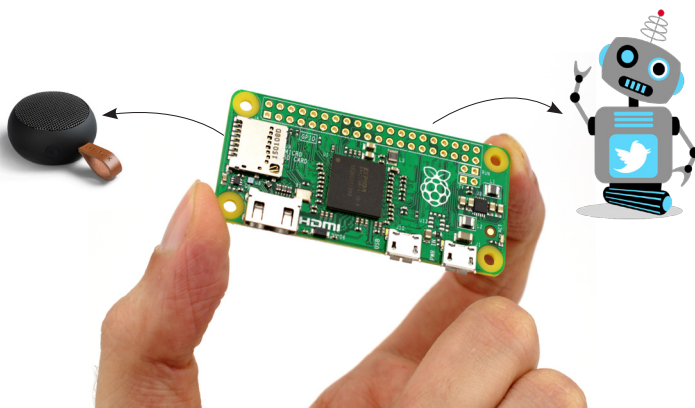




Have Some Fun Doing These Useful Home IoT Projects With Raspberry Pi

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IoT projects can be fun and a great learning experience for budding programmers. Due to the pandemic, most of us are still stuck at home. This can be a great opportunity to dive into some DIY projects to enhance our programming and tech skills. In this article, we look at a few useful IoT projects that are fun and can be done from the comfort of your home using a Raspberry Pi along with tools and gadgets that pretty much everyone owns



Internet of Things (IoT) is defined as devices connected over the Internet that are able to communicate or send data between each other. IoT is a very broad area and its capabilities are limitless.

A range of skills is needed to employ and actualise IoT technologies, and most of these skills are hard to come by as the technology is still maturing and has a lot of vulnerabilities. Having said that, learning how to do simple DIY projects using IoT is a great place to start for any technology enthusiast, specially programmers.

Raspberry Pi

The Raspberry Pi is a small, low-cost computer that can be used for, among other functions, DIY (do-it-yourself) IoT projects. It is a plug and play computer with an open source operating system named Raspbian, which is a distribution of Linux. It was initially designed to help kids at school learn the basics of programming at a low cost.

However, since then, Raspberry Pi has become a favourite of many open source and IoT enthusiasts around the world, who have pushed the functionality and usability of this device to great extremes. Added to that, Raspberry Pi also has a huge and active community online to help answer

your questions and offer support.

So, without further ado, let us jump in and discuss three different fun-filled and useful DIY projects that you can do today in the comfort of your home, just by using a Raspberry Pi and other simple tools and gadgets commonly owned by most.

Domoticz

Domoticz is a home automation system that lets you monitor and configure various devices like lights, switches, and other sensors and meters. By using it, you can easily manage your home devices with your mobile phone. Domoticz can be installed and used on Raspberry Pi. Let us take a look at how this is done.

What you need.

- Power supply and source
- MicroSD card
- Raspberry Pi
- Ethernet cable
- Microphone

Note. Set up the Raspberry on a static IP address.

(i) Upgrade all the packages on the Raspberry Pi by running the following commands:

```
sudo apt update  
sudo apt upgrade
```

(ii) Install Domoticz in the Raspberry Pi by running the install script

from its website. Run the following command:

```
curl -L https://install.domoticz.com | bash
```

After the script run completes, press the *Enter* key.

You will then choose the services you want Domoticz to operate on. For this tutorial, we will leave both HTTP and HTTPS on. But some people prefer to disable the HTTP so that they can use the more secure HTTPS. Press *Enter* to continue.

Choose the port you want HTTP to run on, although this is usually by default 8080; so just press *Enter*. And do the same for the HTTPS port, which is 443 by default.

(iii) Choose the Domoticz installation folder, which usually by default is the *home* folder. Press *Enter*.

(iv) The installation process is complete at this point and you will be shown two IP addresses, which you should take note of. Press *Enter* to complete the installation process.

Configuration of the Domoticz Web interface. Get the static IP address we assigned to the Raspberry Pi above. Through a Web browser, run:

```
http://[IPADDRESS]:8080
```

At *IPADDRESS* use the static Raspberry IP address.

At this point you will have access to the Domoticz Web interface and can